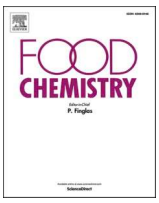
Food Chemistry 493 (2025) 145681



Contents lists available at ScienceDirect

Food Chemistry

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DMU-SPME variable temperature extraction: Revealing the flavor characteristics of strong-aroma baijiu by volatilomics

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ARTICLE INFO

Keywords:
Dual-mode unity solid-phase microextraction
Volatilomics
Variable temperature
Extraction
Strong-aroma baijiu

ABSTRACT

Comprehensive determination of volatilomics in strong-aroma Baijiu, including volatility and low-volatility, poses significant challenge but is crucial for volatilomics research in Baijiu and other spirits. This study developed a dual-mode unity solid-phase microextraction (DMU-SPME) approach combined with variable-temperature extraction (involving NaCl addition and fiber immersion depth optimization) and gas chromatography–mass spectrometry (GC–MS) to establish a rapid and simultaneous method for determining both volatility and low-volatility in strong-aroma Baijiu. DMU-SPME with variable-temperature extraction identified 103 substances, exceeding headspace SPME (HS-SPME) by 32 and immersion SPME (DI-SPME) by 25, while uniquely extracting 19 compounds undetectable by either method. Analysis of three strong-aroma Baijiu samples revealed 46 common characteristic flavor compounds (11 with RI > 1560) and 36 unique flavor compounds (VIP > 1.0; 11 with RI > 1560). DMU-SPME combined with variable-temperature extraction demonstrated superior efficiency for volatilomics, enabling enhanced equilibrium and adsorption of diverse aroma compounds in Baijiu and other spirits.

1. Introduction

Strong-aroma Baijiu is a traditional distilled liquor with a long history (Zhang et al., 2021) and is known for its rich cellar fragrance, sweet and cleansing notes, harmonious bouquet, and long-lasting aftertaste (Yan et al., 2015). This distinctive flavor arises from a complex mix of trace components, including volatility compounds, low-volatility and non-volatility substances, such as fatty acids, all contributing to Baijiu's unique taste and strong aroma (Liu et al., 2022). Luzhou Laojiao liquor, brewed using traditional solid-state fermentation processes, is a typical representative of the strong-aroma type of Baijiu (Chen et al., 2024).

Current pretreatment methods for analysing Baijiu aroma components include direct injection, liquid-liquid extraction (LLE), stir bar sorptive extraction (SBSE), simultaneous distillation extraction (SDE), solvent-assisted flavor evaporation (SAFE), solid-phase extraction (SPE) and solid-phase microextraction (SPME) (Hong et al., 2021; Li et al., 2023; Merkle et al., 2015; Ruvalcaba et al., 2019; Wang et al., 2022). Direct injection is simple and quick, reducing sample loss but has low sensitivity for the detection of low-volatile substances (Mirnaghi et al., 2013). LLE is cost-effective, with high recovery rates, but requires large sample volumes and significant amounts of organic reagents (Cortada et al., 2011). SDE and SAFE have shown some effectiveness in practical applications; however, these methods are prone to introducing co-extractants, which may affect the purity of the extract (Duru et al., 2002). SBSE demonstrates excellent sensitivity and selectivity without the need for organic solvents, although it is characterized by prolonged extraction time and significant background residue (David & Sandra, 2007). SPE is highly efficient and selective, but the target compounds appear to be inadequately retained on the adsorbent and incompletely eluted during the extraction process (Le Bizec et al., 2009). SPME is an efficient, rapid, solvent-free technique, and HS-SPME is by far the most common form of SPME (Mirnaghi et al., 2013; Kataoka et al., 2000). However, HS-SPME suffers from incomplete extraction and competitive adsorption effects, leading to lower sensitivity for semi-volatile and low-volatile components in Baijiu (Merkle et al., 2015). Therefore, developing a comprehensive extraction method for rapid, simultaneous

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https://doi.org/10.1016/j.foodchem.2025.145681
Received 13 March 2025; Received in revised form 13 July 2025; Accepted 22 July 2025
Available online 23 July 2025
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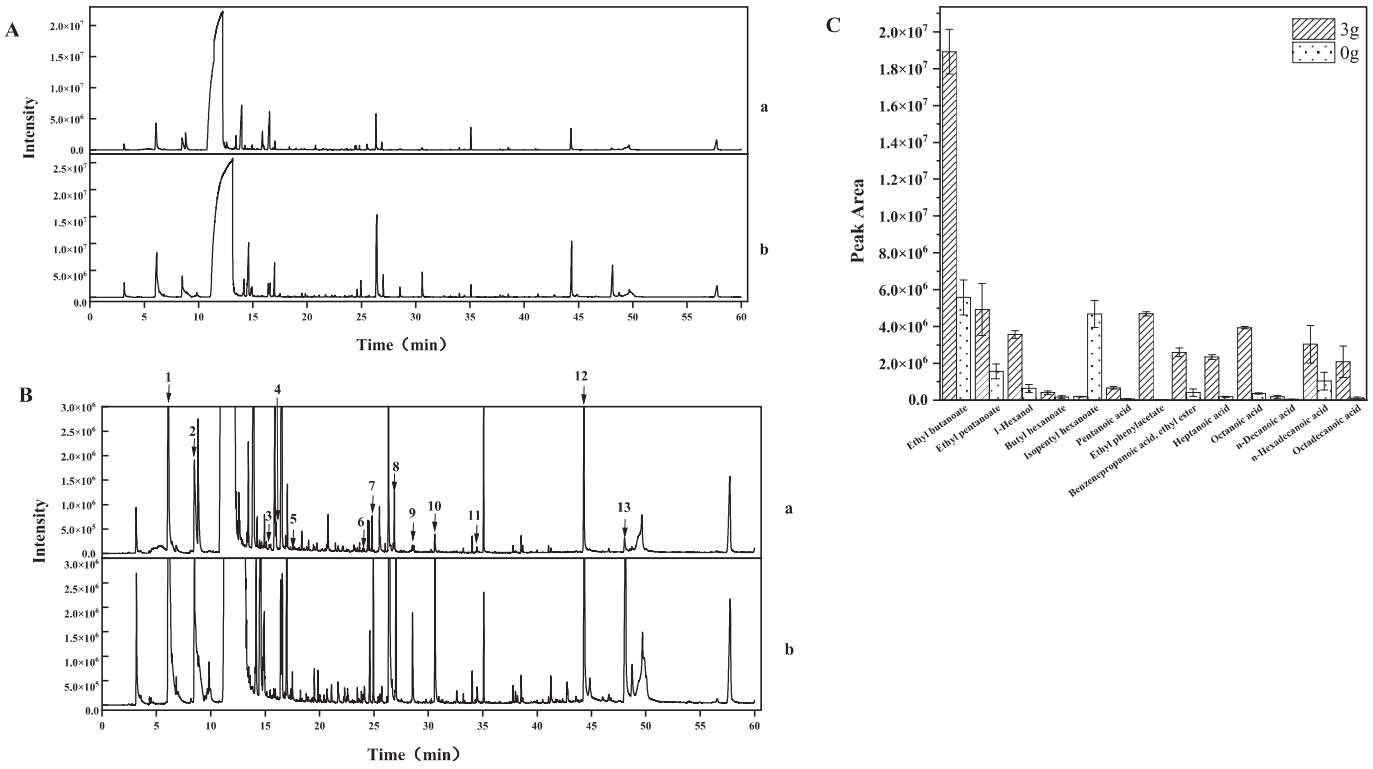


Fig. 1. (A) Overall comparison of chromatograms of Baijiu with different salt concentrations: (a) no salt, (b) 3 g salt. (B) Enlarge comparison of chromatograms of Baijiu with different salt concentrations: (a) no salt, (b) 3 g salt. 1)- Ethyl butanoate, 2)- Ethyl pentanoate, 3)- 1-Hexanol, 4)- Butyl hexanoate, 5)- Isopentyl hexanoate, 6)- Pentanoic acid, 7)- Ethyl phenylacetate, 8)- Benzenepropanoic acid, ethyl ester, 9)- Heptanoic acid, 10)- Octanoic acid, 11)- n-Decanoic acid, 12)- n-Hexadecanoic acid, 13)- Octadecanoic acid. (C) The effect of salt concentration on extraction capacity of the 13 key compounds of Baijiu identified through the literature review with DMU-SPME.

extraction of volatility and low-volatility compounds in strong-aroma Baijiu is essential.

The dual-mode unity solid-phase microextraction (DMU-SPME) technique combines headspace solid-phase microextraction (HS-SPME) and direct immersion solid-phase microextraction (DI-SPME), enabling the efficient extraction of both volatile and low-volatile organic compounds (Ma, Zhang, et al., 2024). Previous studies have shown that DMU-SPME extracted 107 compounds from soy sauce, covering the detection ranges of both HS-SPME and DI-SPME, including 16 fatty acids (C_2–C_{18}) in soy sauce and balsamic vinegar matrices (Ma, Li, et al., 2024, Ma, Zhang, et al., 2024). However, dynamic linked position unity solid-phase microextraction (DLPU-SPME) revealed competitive adsorption between the initial HS-SPME and the subsequent DI-SPME extraction stage (Ma et al., 2023). Critically, temperature governs SPME kinetics: higher temperatures favor volatile compound extraction, while lower temperatures enhance low-volatility analyte recovery (Câmara et al., 2007; Risticevic et al., 2010).

We therefore hypothesize that DMU-SPME coupled with optimized variable-temperature extraction (including adjustments to immersion depth and salt addition) mitigates competitive adsorption and enhance the detection of low-volatility and semi-volatile compounds. This study aims to develop and validate a rapid and simultaneous method for determining both volatility and low-volatility compounds in strong-aroma Baijiu, suitable for quality assessment and volatilome characterization applications in Baijiu and other spirits.

2. Materials and methods

2.1. Materials and reagents

Ethanol (GC grade) was obtained from Aladdin (Shanghai, China). A C_7–C_{30} n-alkane mixture (Sigma-Aldrich, Shanghai, China) was used to determine the linear retention indices (RIs). 2-Octanol (purity >99 %) was purchased from Macklin Biochemical Co. (Shanghai, China). NaCl was obtained from Tianda Chemical Experimental Factory (Tianjin, China). Ultrapure water was used in all experiments. A hydrophilic-lipophilic balance/polydimethylsiloxane fiber (HLB/PDMS, 2 cm, 75 μm film thickness, Fangtianju®, Reyummy, China) was used for DMU-SPME extractions. Three commercial strong-aroma Baijiu samples representing different grades/ages were analyzed: National Cellar 1573 (GJ; high-end), Luzhou Laojiao Tequ (TQ; mid-high-end), and Luzhou Laojiao Old Touqu (8NJ, mid-low-end). Samples were purchased from local supermarkets and stored at 4 °C before analysis. Brand names are mentioned for research purposes only, not for advertising.

2.2. Sample preparation

Samples were diluted to 10 % ethanol (*v*/v) with ultrapure water. The internal standard (2-octanol, 450 μg/mL, 100 μL) was added to 10 % sample (8 mL) into a 20 mL headspace vial. Optimal DMU-SPME conditions were determined by comparing: (1) NaCl treatments (with or without saturated NaCl solution), (2) three temperature protocols—constant extraction (50 °C, 45 min), ambient-variable extraction (50 °C for 30 min followed by 20 °C for 15 min), and low-temperature-variable extraction (50 °C for 30 min followed by 2 °C for 15 min), and (3) fiber immersion depths (0.5, 1.0, and 1.5 cm). A stable ambient temperature of 20 °C was chosen to avoid fluctuations at 25 °C (Kustrin et al., 2023), and 2 °C was selected as the low temperature to prevent ice formation that might occur at 0 °C. Extraction was conducted at a stirring speed of 400 rpm. For comparison, the extraction parameters for HS/DI-SPME were set at 50 °C for 45 min (Zhang et al., 2020).

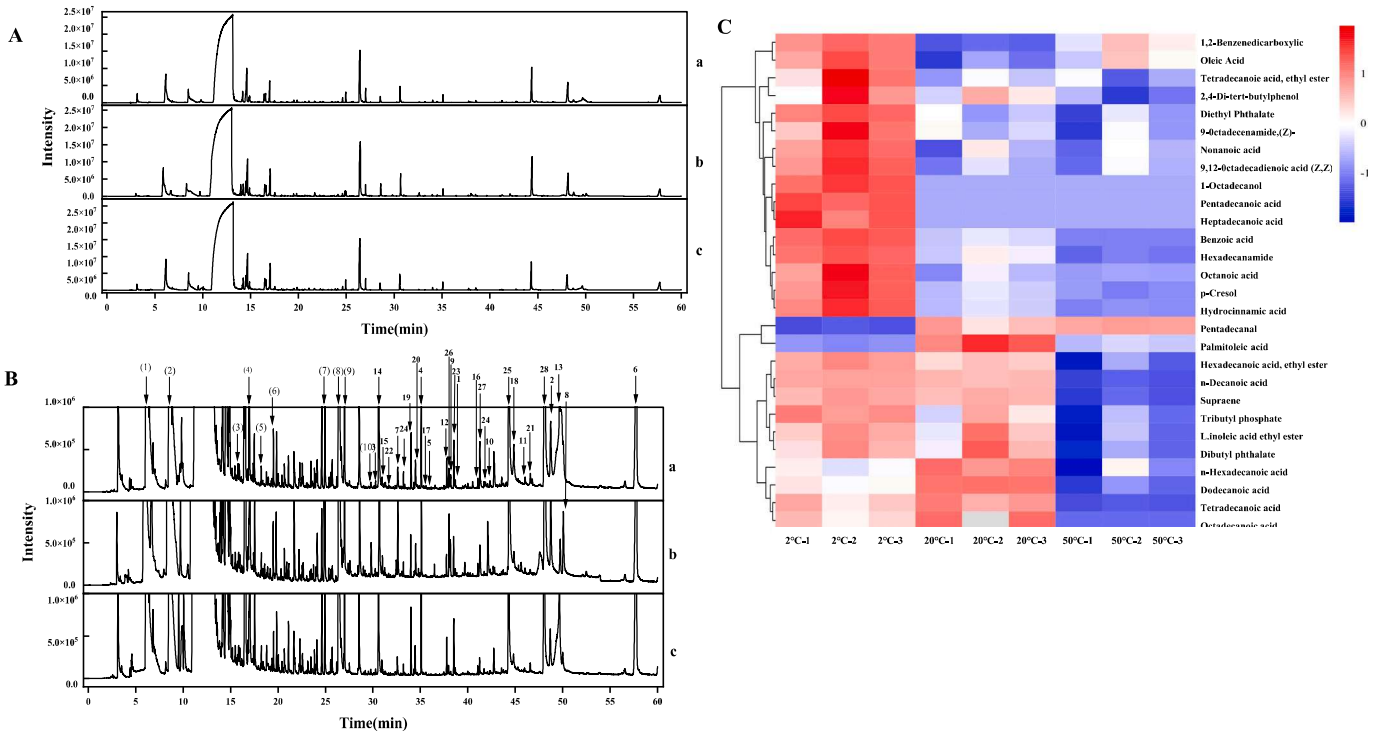


Fig. 2. (A) Overall comparison of chromatograms of Baijiu with three extraction temperatures: (a) constant extraction, (b) ambient-variable extraction, (c) low-temperature-variable extraction. (B) Enlarge comparison of chromatograms of Baijiu with three extraction temperatures: (a) constant extraction, (b) ambient-variable extraction, (c) low-temperature-variable extraction. (1)- Ethyl butyrate, (2)- Ethyl valerate, (3)- Nonanal, (4)- Heptyl Formate, (5)- Benzaldehyde, (6)- Longifolene, (7)- Ethyl phenylacetate, (8)- Hexanoic acid, (9)- Ethyl 3-phenylpropionate, (10)- Butyl phenylacetate. (C) The peak areas of the substances analyzed at 30–60 min for the three different extraction temperatures with a heatmap plot. The most and least intense relative abundance values are depicted in red and purple, respectively. 1)- 1,2-Benzenedicarboxylic, 2)- Oleic Acid, 3)- Tetradecanoic acid, ethyl ester, 4)- 2,4-Di-tert-butylphenol, 5)- Diethyl Phthalate, 6)- 9-0ctadecenamide, (Z)-, 7)- Nonanoic acid, 8)- 9,12-0ctadecadienoic acid (Z,Z), 9)- 1-Octadecanol, 10)- Pentadecanoic acid, 11)- Heptadecanoic acid, 12)- Benzoie acid, 13)- Hexadecanamide, 14)- Octanoic acid, 15)- p-Cresol, 16)- Hydrocinnamic acid, 17)- Pentadecanal, 18)- Palmitoleic acid, 19)- Hexadecanoic acid, ethyl ester, 20)- n-Decanoic acid, 21)- Supraene, 22)- Tributyl phosphate, 23)- Linoleic acid ethyl ester, 24)- Dibutyl phthalate, 25)- n-Hexadecanoic acid, 26)- Dodecanoic acid, 27)- Tetradecanoic acid, 28)- Octadecanoic acid. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

2.3. SPME procedure

Before sample analysis, the HLB/PDMS fiber was pretreated by inserting it directly into the GC inlet at 250 °C for 30 min to desorb any potential interfering substances. The sample was then equilibrated at 50 °C in a thermostatic bath for 10 min. Following extraction, the SPME fiber was removed from the headspace vial, rinsed with ultrapure water for 30 s, and gently wiped with lint-free paper to remove excess water (Ma, Zhang, et al., 2024). After that, the fiber was inserted into the injection port of GC (250 °C) to desorb the analytes.

2.4. Instrumental analysis method for GC–MS

GC–MS analysis was performed using an Agilent 7890B/5977B system (Agilent Technologies Inc., California, USA) equipped with a polar VF-WAXms column (30 m × 0.25 mm × 0.25 μm) to separate and detect the compounds extracted from the samples. Instrument parameters were based on a previous method (Ma, Zhang, et al., 2024). The GC oven was initially set to 35 °C for 3 min, then ramped at 5 °C/min to 250 °C and held for 14 min. The injection was non-split, with the injection port temperature set to 250 °C. Helium was used as the carrier gas at a flow rate of 1 mL/min. Mass spectrometric analysis was conducted on an Agilent 5977B instrument, operating in scan mode with a range of 50–550 *m/z*. The ion source was set to 230 °C with electron impact ionization at 70 eV.

2.5. Statistical analysis

All experiments were performed in triplicate. GC–MS data were analyzed using Mass Hunter Workstation Qualitative Analysis software (version B.10.00, Agilent Technologies, Inc. 2016, Santa Clara, CA, USA). Orthogonal Partial Least Squares Discriminant Analysis (OPLS-DA) was performed using MetaboAnalyst 5.0. A hot spot was used for heat map visualization analysis. A Venn plot was constructed using the TBtools software.

3. Results and discussion

3.1. Effect of salt concentration on extraction capacity of DMU-SPME

The addition of salt plays a critical role in the extraction efficiency of DMU-SPME due to the salting-out effect (Liu et al., 2018). As shown in Fig. 1A and B, the addition of salt increased the peak intensity of some aromatic compounds, such as ethyl butanoate and ethyl pentanoate (Fig. 1C), indicating that the extraction of most volatile organic compounds increased significantly with the addition of NaCl (Rocha et al., 2001).

However, compounds such as isopentyl hexanoate showed a slight decrease in peak intensity when NaCl was added (Fig. 1C). This could be because NaCl reduced the solubility of these compounds and affected their competitive uptake by the SPME fibers (Fiorini et al., 2015). Although the extraction of some compounds decreased slightly, the effectiveness of DMU-SPME for Baijiu flavor analysis remained

Table 1
Comparison of peak areas for ten characteristic flavor compounds at three different extraction temperatures. Each value is the mean ± SD of triplicate measurements ($n = 3$).

RT	Compounds	Constant extraction	Ambient-variable extraction	Low-temperature-variable extraction
5.889	Ethyl butyrate	19,296,754.67 ± 413,936.68[a]	19,496,160.67 ± 427,606.03[a]	19,828,952.33 ± 54,543.53[a]
8.316	Ethyl valerate	4,635,973.67 ± 19,867.50[a]	4,655,800.33 ± 16,475.21[a]	4,664,878.33 ± 21,198.09[a]
15.728	Nonanal	57,949.33 ± 994.19[a]	58,413.00 ± 46.89[a]	57,674.00 ± 289.02[a]
17.387	Heptyl Formate	186,795.33 ± 1066.18[a]	185,500.00 ± 3100.33[a]	188,656.00 ± 2802.05[a]
18.915	Benzaldehyde	33,433.33 ± 400.28[a]	33,497.67 ± 268.43[a]	32,564.33 ± 917.39[a]
19.71	Longifolene	17,758.67 ± 124.07[a]	17,451.33 ± 369.39[a]	17,758.00 ± 82.02[a]
24.88	Ethyl phenylacetate	4,654,739.67 ± 34,549.14[a]	4,678,766.00 ± 30,203.39[a]	4,560,661.67 ± 38,937.96[a]
26.46	Hexanoic acid	38,585,858.00 ± 337,714.97[a]	38,789,288.67 ± 186,120.94[a]	38,243,124.33 ± 955,691.50[a]
27.032	Ethyl 3-phenylpropionate	2,543,743.00 ± 20,193.98[a]	2,556,251.33 ± 27,004.62[a]	2563,043.33 ± 22,654.28[a]
28.494	Butyl phenylacetate	93,407.00 ± 720.40[a]	93,566.33 ± 904.75[a]	92,931.33 ± 1225.22[a]

There is no significant difference (P > 0.05) between the date marked with the same letters.

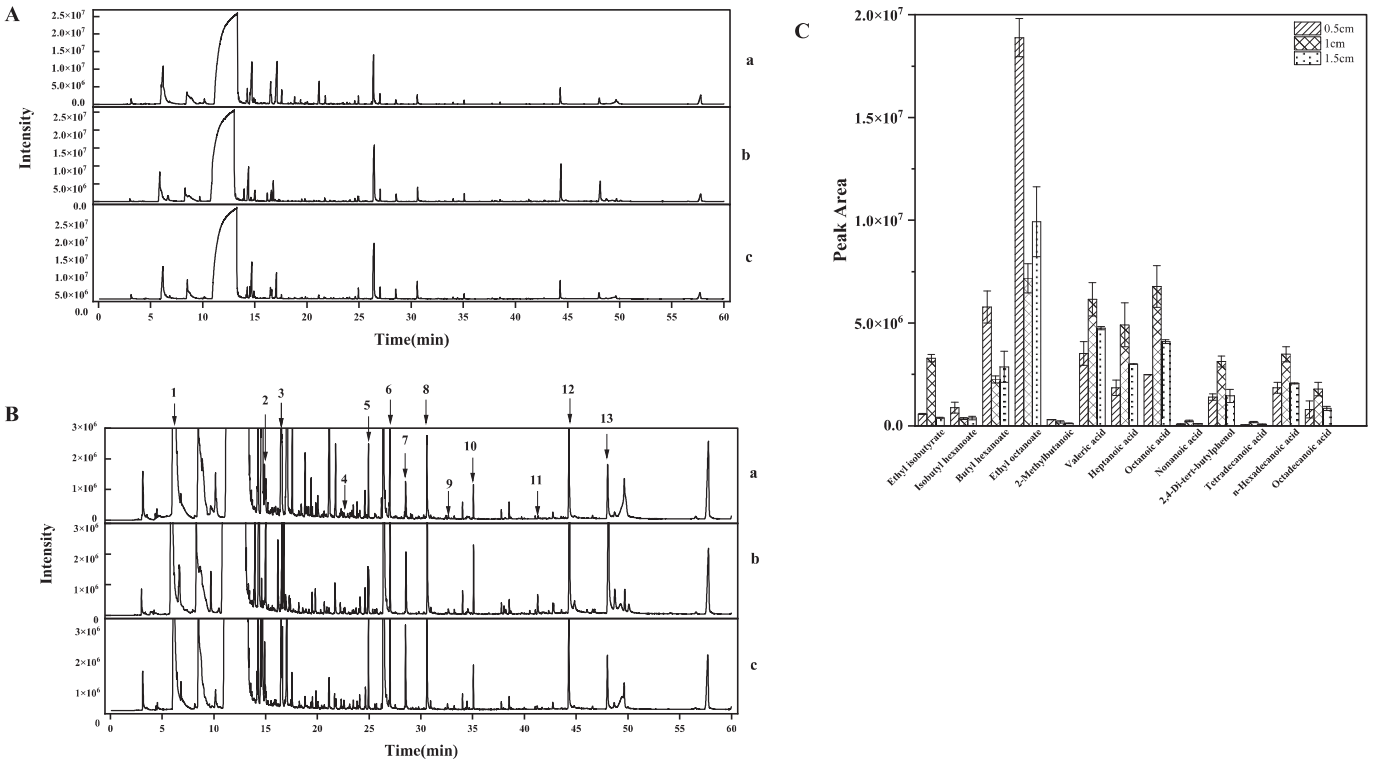


Fig. 3. (A) Overall comparison of chromatograms of Baijiu with different immersion lengths: (a) 0.5 cm, (b) 1.0 cm, (c) 1.5 cm. (B) Enlarge comparison of chromatograms of Baijiu with different immersion lengths: (a) 0.5 cm, (b) 1.0 cm, (c) 1.5 cm. (C) The effect of length in liquid on extraction capacity of the 13 key compounds of Baijiu identified through the literature review with DMU-SPME. 1)- Ethyl isobutyrate, 2)- Isobutyl hexanoate, 3)- Butyl hexanoate, 4)- Ethyl octanoate, 5)- Ethyl phenylacetate, 6)-Benzenepropanoic acid,ethyl, 7)- Heptanoic acid, 8)- Octanoic acid, 9)- Nonanoic acid, 10)- 2,4-Di-tert-butylphenol, 11)- Tetradecanoic acid, 12)- n-Hexadecanoic acid, 13)- Octadecanoic acid.

unaffected. Therefore, NaCl addition was retained for better extraction of flavor compounds in Baijiu in this study.

3.2. Effect of temperature on extraction capacity of DMU-SPME

The extraction temperature affects the diffusivity of aromatic substances and thus can significantly influence the extraction efficiency of DMU-SPME (Rocha et al., 2001). Chromatograms analysis (Fig. 2A and B) revealed distinct variations in peak intensities across three extraction temperature protocols, confirming temperature-dependent effects on volatile and low-volatile compound recovery. Analysis of peak areas for 10 key flavor substances (Chen et al., 2024) during the initial 30 min showed no significant differences ($P > 0.05$) compared to literature-reported markers of strong-aroma Baijiu (Table 1), suggesting negligible temperature effects in this early extraction phase. However, the heat map analysis (Fig. 2C) demonstrated substantially higher compound recovery during the 30–60 min elution under low-temperature-variable extraction compared to both constant and ambient-variable protocols. This method yielded optimal results, uniquely detecting three additional compounds: stearyl alcohol, pentadecanoic acid, and heptadecanoic acid. Furthermore, peak intensities of 13 low-volatility compounds—including nine long-chain fatty acids (e.g., oleic acid) and four long-chain fatty acid esters (e.g., ethyl palmitate)—were significantly enhanced under low-temperature conditions, surpassing the performance of constant extraction.

This performance optimization stems from DMU-SPME's ability to balance volatility and solubility: higher temperatures favor volatile compound extraction, while lower temperatures enhance recovery of semi-volatile species (Câmara et al., 2007). Kinetically, elevated temperatures increase analyte diffusion coefficients and headspace capacity, accelerating extraction rates. Thermodynamically, however, they reduce the coating/matrix partition coefficients for many analytes. To

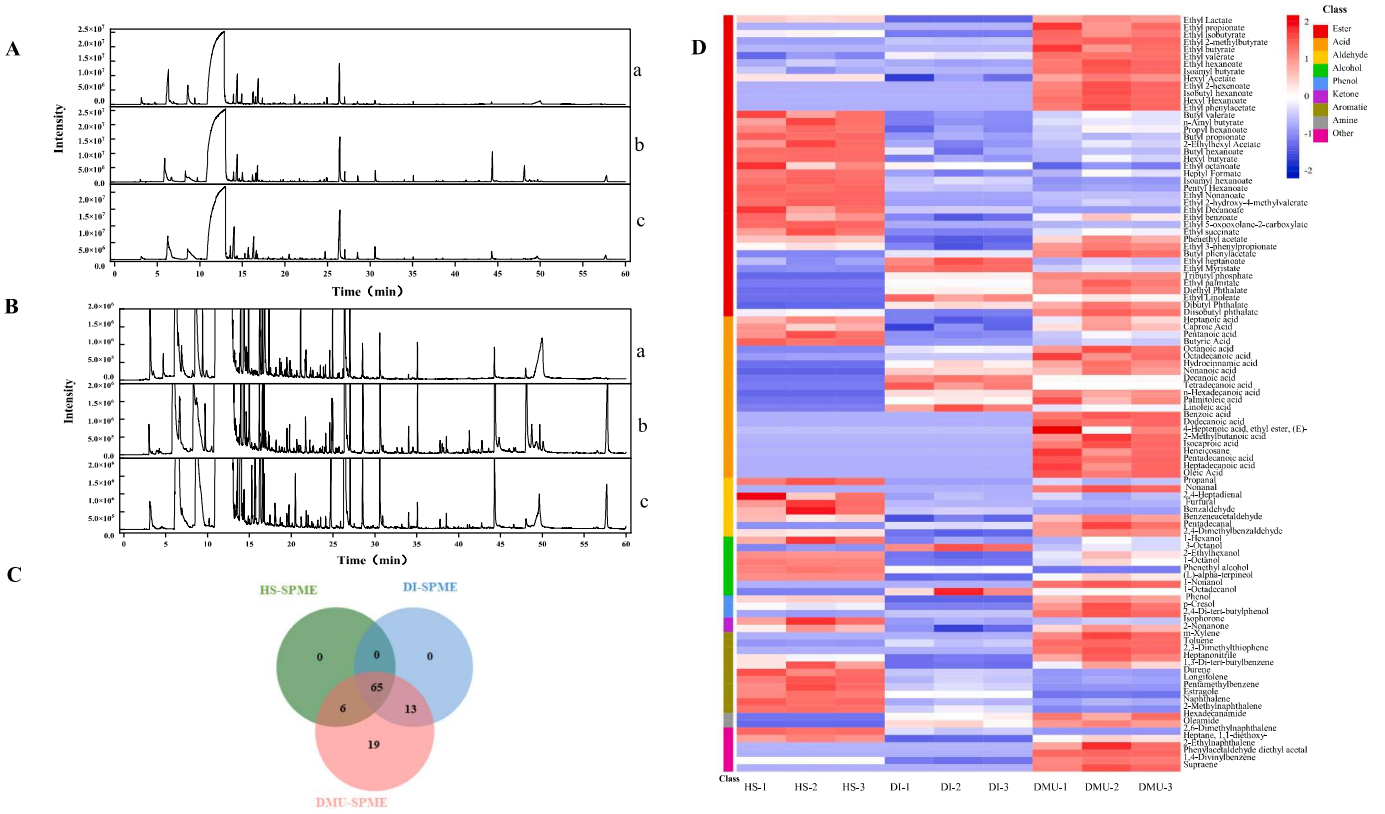


Fig. 4. (A) Overall comparison of chromatograms of Baijiu with three extraction methods: (a) HS-SPME, (b) DMU-SPME, (c) DI-SPME. (B) Enlarge comparison of chromatograms of Baijiu with three extraction methods: (a) HS-SPME, (b) DMU-SPME, (c) DI-SPME. (C) Venn diagram of the number of compounds extracted by HS-SPME, DMU-SPME and DI-SPME. (D) Heatmap of the peak areas for the substances analyzed by HS-SPME, DMU-SPME and DI-SPME. The most and least intense relative abundance values are depicted in red and purple, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

maximize coverage, we implemented a novel two-stage protocol: 50 °C for 30 min (targeting volatiles) followed by 2 °C for 15 min (targeting low-volatility compounds). This sequential approach leverages kinetic advantages at higher temperature while mitigating thermodynamic limitations at lower temperature.

3.3. *Effect of fiber's immersion depth on extraction capacity of DMU-SPME*

The depth of fiber immersion in a solid-liquid-gas system during DI-SPME phase is a key factor affecting the analyte extraction efficiency (Ma, Zhang, et al., 2024). Fig. 3A and B show the effect of immersion depth on the extraction of Baijiu flavor compounds. A fiber immersion depth of 1.0 cm resulted in higher peak intensities for some aroma compounds within 30–60 min compared with other immersion lengths. As the immersion depth increased, the fiber surface area in contact with the Baijiu increased, facilitating the adsorption of more analytes. However, excessive immersion may lead to fiber saturation, reducing its ability to effectively capture volatile compounds (Salum et al., 2017).

Fig. 3C shows that the peak intensities of some volatile compounds such as isobutyl hexanoate were lower at 1.0 cm than at an immersion length of 0.5 cm. However, the peak intensities of low-volatile fatty acids such as heptanoic acid, octanoic acid, and nonanoic acid were much higher than those obtained at depths of 0.5 cm and 1.5 cm. The distribution of the analytes during extraction can be influenced by the volatility of the compounds and the affinity of the fibers for specific classes of compounds (Yang et al., 2013). Considering the overall extraction efficiency for volatile and low-volatility compounds, an immersion depth of 1.0 cm provided the best balance between headspace and liquid phase access. Therefore, an immersion depth of 1.0 cm was optimal for effectively extracting Baijiu aromatic substances.

3.4. *Comparison of volatile compounds in samples under three extraction methods*

The extraction performances of DMU-SPME, HS-SPME, and DI-SPME for Baijiu flavor substances were compared using GC–MS. Representative chromatograms are shown in Fig. 4A and B. Compared to HS-SPME and DI-SPME, DMU-SPME demonstrated a significant advantage by extracting a greater number of compounds. Fig. 4C presents a Venn

Table 2
Comparison of the DMU-SPME method with other techniques.

	Esters	Acids	Alcohols	Aldehyde	Substance Total	Reference
HS-SPME	32	6	5	6	71	This work
DI-SPME	40	22	8	8	78	This work
DMU-SPME	33	13	6	6	103	This work
Liquid-liquid	32	10	13	7	78	Hong et al., 2023
Direct Sampling	14	9	7	1	35	Hong et al., 2021
HS-SPME	23	7	13	4	56	Fan et al.2005

Table 3
Unique Aroma Compounds Identified in DMU-SPME Variable Temperature Extraction of Strong-Aroma Baijiu.

	Compound	RT	RI	Sensory characteristic	Reference
Esters (7)					
1	Ethyl isobutyrate	4.232	960	Apple, pear	Zhu et al., 2025
2	Ethyl 2-hexenoate	14.620	1357	fruity, pineapple,	Chen et al., 2024
3	Isobutyl hexanoate	14.720	1356	pineapple, green	Chen et al., 2024
4	4-Heptenoic acid ethyl ester (*E*)-	15.272	1377	unknown	
5	2-hydroxy-4-methylvalerate	18.819	1515	contributes to flavor in the mouth	Yamazaki et al., 2024
6	Hexyl hexanoate	21.126	1611	herbal, fresh, vegetable	Chen et al., 2024
7	Ethyl phenylacetate	24.880	1754	sweet, floral, honey, rose	Chen et al., 2024
Acids (8)					
1	2-Methylbutanoic	22.640	1674	cheese	Tachihara et al., 2006
2	Isocaproic acid	25.519	1813	Vinegar-like	Zhang et al., 2025
3	Undecanoic Acid	36.558	2398	rancid, fatty	Li et al., 2023
4	Benzoic Acid	37.665	2457	unknown	
5	Dodecanoic acid	38.036	2490	Unpleasant, animal, leather	Deuscher et al., 2020
6	Pentadecanoic Acid	42.346	2779	waxy	Miao et al., 2022
7	Heptadecanoic acid	46.048	3027	unknown	
8	Oleic Acid	48.732	3150	Modulating the Volatility Characteristics of Esters	Liu et al., 2024
Aldehydes (2)					
1	Nonanal	15.728	1392	rose, fresh	Chen et al., 2024
2	Phenylacetaldehyde diethyl acetal	23.338	1711	honey-like	Monforte et al., 2021
Alcohols (1)	1-Nonanol	22.142	1658	fatty, floral, rose	Chen et al., 2024
Hydrocarbon (1)	m-Xylene	8.235	1123	unknown	

diagram comparing the number of compounds extracted by each method; HS-SPME identified 71 compounds, DI-SPME identified 78, while DMU-SPME extracted 103 compounds, encompassing all those extracted by the other two. This confirms that DMU-SPME effectively combines the extraction capabilities of HS-SPME and DI-SPME (Ma, Zhang, et al., 2024), enabling efficient simultaneous extraction of both volatile and low-volatility organic compounds. A heatmap of the peak areas for the substances detected by the three different extraction methods further confirmed these results (Fig. 4D). DMU-SPME's superior efficiency in Baijiu flavor analysis stems from its variable temperature extraction mode, which enables prolonged low-temperature interaction favoring low-volatility analytes. This enhances adsorption of a wider range of compounds and broadens the spectrum of detectable compounds beyond what is achievable with single HS-SPME or DI-SPME modes, highlighting its broad extraction range.

Additionally, the performance of DMU-SPME combined with variable temperature extraction was compared with reported results using traditional methods (Table 2). Liquid-liquid extraction identified 78 aroma compounds in strong-flavoured Baijiu (Hong et al., 2023), whereas direct sampling detected 35 trace components in Luzhou Laojiao (Hong et al., 2021). HS-SPME analysis of Yanghe Daqu wine identified 56 compounds (Fan & Qian, 2005). Compared with these methods, DMU-SPME extraction demonstrated significant advantages, extracting a broader range of compounds.

DMU-SPME demonstrated excellent performance in extracting esters and acids from Baijiu. DMU-SPME extracted 40 esters, exceeding the totals obtained by HS-SPME (32 esters) and DI-SPME (33 esters). DMU-SPME extracted 22 organic acids, significantly more than HS-SPME (6 acids) or DI-SPME (13 acids). Comparing the results of this study with those of other peer studies on important acids and esters reveals: liquid-liquid extraction identified 32 esters and 10 acids in concentrated Baijiu (Hong et al., 2023), whereas direct sampling detected 14 esters and 9 acids in Luzhou Laojiao (Hong et al., 2021). In Yanghe Daqu wine, HS-SPME identified 23 esters and 7 acids (Fan & Qian, 2005). Importantly, DMU-SPME extracted 19 more flavor compounds (Table 3) that were not detected by either HS-SPME and DI-SPME alone (7 esters, 8 acids, 1 alcohol, 2 aldehydes, 1 hydrocarbons). These additional compounds serve as critical flavor contributors, with their sensory roles—particularly in enhancing aroma persistence and complexity in Baijiu— are summarized in Table 3.

These comparisons underscore the superior efficiency of DMU-SPME in extracting a broader spectrum of aroma compounds, highlighting its advantages for Baijiu flavor analysis, primarily due to two synergistic mechanisms: (1) Variable-Temperature Optimization: The two-stage protocol (from 50 °C /30 min to 2 °C /15 min) leverages kinetic advantages at high temperature for volatiles (e.g., esters like ethyl isobutyrate/apple-pear aroma) while enhancing thermodynamic partitioning at low temperature for semi-volatiles—uniquely detecting compounds like stearyl alcohol and pentadecanoic acid that escape single-mode SPME. (2) Dual-Mode Integration: Simultaneous HS/DI extraction overcomes inherent limitations of individual techniques: HS-SPME captures gas-phase volatiles but misses low-volatility species (e.g., long-chain esters like ethyl nonanoate critical for Baijiu's persistent aroma), while DI-SPME targets liquid-phase low-volatiles (e.g., waxy pentadecanoic acid) but overlooks highly volatile analytes.

3.5. Analysis of aroma components of Luzhou-flavor Baijiu

To investigate the aroma characteristics of strong-aroma Baijiu, the composition and relative content of aromatic compounds in the three strong-flavoured Baijiu samples were analyzed using DMU-SPME combined with GC–MS. Table 4 indicates that 103, 79, and 86 aroma compounds were identified in GJ, TQ, and 8NJ, respectively, with GJ exhibiting the highest number of aromatic compounds. The Venn diagram in Fig. 5A reveals 46 common aroma components across the three Baijiu types, including 22 esters, 12 acids, four aldehydes, two alcohols, one aromatic compound, one ketone, one phenolic compound, and three others.

Compared to literature reports using conventional methods (Chen et al., 2024), the lower relative content of some short-chain esters (e.g., ethyl acetate, ethyl butyrate, ethyl valerate) in our DMU-SPME results is attributed to the unique design of DMU-SPME. First, dual-mode priority: Unlike HS-SPME (which focuses on the gas phase), the combined headspace/direct immersion mode of DMU-SPME favors the extraction of medium and long-chain esters, leading to reduced recovery of short-chain esters. Second, temperature-induced loss: Cooling from 50 °C to 2 °C during the extraction causes partial recondensation of volatilized esters into the liquid phase, thereby decreasing their capture by the fiber. Third, competitive adsorption: Hexanoic acid and other long-chain fatty acids and fatty acid ester compete for binding sites on HLB/PDMS through its polar carboxyl group, potentially suppressing the extraction of some esters such as ethyl butyrate. Therefore, DMU-SPME inherently prioritizes less volatile compounds, resulting in a systematic underestimation of short-chain esters compared to HS-SPME.

Table 4
The relative concentration of aroma compounds in GJ,8NJ and TQ. Each value is the mean ± SD of triplicate measurements (n = 3).

		RI		MS						Relative Concentration (mg L^{-1})[e]		
Peak number	RT (min)	RIref[a]	RIcal[b]	Mass	ion[c]	Formula	Compounds[d]	CAS	GJ	TQ	8NJ	Identification
Ester												
1	3.122		885	70	61	C4H8O2	Ethyl acetate	141–78-6	ND	6.80 ± 0.70	8.36 ± 0.85	MS
4	4.087	954	951	75	57	C5 H10 O2	Ethyl propionate	105–37-3	0.47 ± 0.18	2.20 ± 0.11	0.59 ± 0.02	MS, RI
5	4.232	962	960	116	71	C6 H12 O2	Ethyl isobutyrate	97–62-1	0.55 ± 0.12	1.65 ± 0.09	0.84 ± 0.07	MS, RI
8	5.889	1038	1035	88	71	C6 H12 O2	Ethyl butyrate	105–54-4	46.87 ± 10.38	ND	ND	MS, RI
9	6.192	1050	1048	102	57	C7 H14 O2	Ethyl 2-methylbutyrate	7452–79-1	0.58 ± 0.10	1.29 ± 0.22	0.68 ± 0.02	MS, RI
10	6.646	1068	1067	88	71	C6 H12 O2	Ethyl butyrate	105–54-4	3.83 ± 1.44	ND	ND	MS, RI
11	6.834	1074	1071	88	85	C7 H14 O2	Ethyl isovalerate	108–64-5	ND	9.96 ± 0.67	0.84 ± 0.09	MS, RI
15	8.316	1129	1131	85	88	C7 H14 O2	Ethyl valerate	539–82-2	8.35 ± 1.15	21.26 ± 2.03	11.22 ± 0.57	MS, RI
16	9.782	1180	1177	87	74	C7H14O2	Methyl hexanoate	106–70-7	ND	ND	0.09 ± 0.01	MS, RI
17	13.024	1293	1270	99	88	C8H16O2	Ethyl hexanoate	123–66-0	1477.82 ± 65.52	3605.30 ± 243.86	2077.50 ± 183.79	MS, RI
18	13.087	1295	1283	55	71	C9 H18 O2	Isoamyl butyrate	106–27-4	2.81 ± 0.17	ND	ND	MS, RI
19	13.150	1297	1303	84	56	C8 H16 O2	Hexyl acetate	142–92-7	1.12 ± 0.12	ND	0.80 ± 0.10	MS, RI
21	13.868	1325	1336	103	85	C9 H18 O2	Butyl valerate	591–68-4	0.84 ± 0.05	2.19 ± 0.30	0.45 ± 0.05	MS, RI
22	13.903	1327	1319	89	71	C9 H18 O2	n-Amyl butyrate	540–18-1	1.63 ± 0.11	ND	ND	MS, RI
23	13.972	1329	1324	117	99	C9 H18 O2	Propyl hexanoate	626–77-7	5.84 ± 0.13	8.55 ± 1.22	4.73 ± 0.23	MS, RI
25	14.397	1346	1341	113	88	C9H18O2	Ethyl heptanoate	106–30-9	33.82 ± 2.58	ND	23.91 ± 1.88	MS, RI
26	14.458	1348	1143	57	69	C7H14O2	Butyl propionate	590–01–2	0.06 ± 0.04	ND	ND	MS, RI
27	14.620	1354	1357	55	97	C8 H14 O2	Ethyl 2-hexenoate	1552-67-6	1.30 ± 0.06	1.68 ± 0.32	0.97 ± 0.06	MS, RI
28	14.720	1358	1356	56	99	C10 H20 O2	Isobutyl hexanoate	105–79-3	0.64 ± 0.05	1.85 ± 0.73	0.18 ± 0.01	MS, RI
31	15.400	1372	1382	53	75	C5H10O3	Ethyl Lactate	97–64-3	0.21 ± 0.03	0.27 ± 0.04	0.22 ± 0.02	MS, RI
32	15.477	1385	1420	57	70	C11 H20 O2	2-Ethylhexyl acetate	103–09-3	0.24 ± 0.01	ND	ND	MS, RI
37	16.396	1420	1418	71	99	C11 H22 O2	2-Methylbutyl hexanoate	2601–13-0	ND	0.48 ± 0.19	ND	MS, RI
38	16.191	1412	1407	99	117	C10 H20 O2	Butyl hexanoate	626–82-4	3.93 ± 0.12	13.51 ± 1.65	4.01 ± 0.45	MS, RI
39	16.25	1414	1419	89	71	C10 H20 O2	Hexyl butyrate	2639-63-6	1.24 ± 0.10	ND	1.16 ± 0.12	MS, RI
41	16.828	1438	1434	131	113	C10 H20 O2	Propyl heptanoate	7778-87-2	ND	0.13 ± 0.02	ND	MS, RI
42	16.776	1436	1435	101	88	C10 H20 O2	Ethyl octanoate	106–32-1	12.71 ± 0.38	69.07 ± 1.93	6.53 ± 0.72	MS, RI
43	17.287	1456	1456	99	70	C11 H22 O2	Isopentyl hexanoate	2198-61-0	1.32 ± 0.06	9.07 ± 0.19	11.20 ± 0.72	MS, RI
45	17.447	1462	1455	70	57	C10H20O2	Heptyl propionate	2216-81-1	ND	0.27 ± 0.03	ND	MS, RI
51	18.255	1493	1507	115	72	C5 H13 N3	Ethylene glycol di-N-butyrate	105–72-6	ND	ND	0.01 ± 0.00	MS, RI
53	18.579	1506	1509	117	70	C11 H22 O2	Pentyl hexanoate	540–07-8	0.22 ± 0.02	2.50 ± 0.07	1.59 ± 0.32	MS, RI
56	19.183	1532	1530	101	88	C11H22O2	Ethyl nonanoate	123–29-5	0.17 ± 0.02	2.83 ± 0.10	ND	MS, RI
60	19.494	1545	1545	87	69	C8 H16 O3	Ethyl 2-hydroxy-4-methylvalerate	10,348–47-7	1.01 ± 0.01	1.37 ± 0.24	0.97 ± 0.05	MS, RI
61	19.692	1553	1552	113	70	C12 H24 O2	3-Methylbutyl heptanoate	109–25-1	ND	ND	0.18 ± 0.02	MS, RI
66	21.126	1612	1611	99	117	C12 H24 O2	Hexyl hexanoate	6378–65-0	0.57 ± 0.03	ND	ND	MS, RI
67	21.324	1621	1622	89	71	C12 H24 O2	Octyl butyrate	110–39-4	ND	0.14 ± 0.02	ND	MS, RI

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Table 4 (*continued*)

		RI		MS						Relative Concentration (mg L^{-1})[e]		
Peak number	RT (min)	RIref[a]	RIcal[b]	Mass	ion[c]	Formula	Compounds[d]	CAS	GJ	TQ	8NJ	Identification
71	21.768	1641	1643	101	88	C12H24O2	Ethyl decanoate	110–38-3	0.16 ± 0.02	5.51 ± 0.73	8.18 ± 0.99	MS, RI
73	22.213	1661	1658	77	105	C9 H10 O2	Ethyl benzoate	93–89-0	0.72 ± 0.01	ND	0.93 ± 0.09	MS, RI
75	22.375	1667	1672	110	105	C12H22O2	Ethyl trans-4-decenoate	76,649–16-6	ND	1.52 ± 0.16	ND	MS, RI
76	22.526	1674	1675	57	101	C7H10O4	Ethyl 5-oxooxolane-2-carboxylate	1126-51-8	0.28 ± 0.04	ND	ND	MS, RI
77	22.549	1675	1675	129	101	C8 H14 O4	Diethyl succinate	123–25-1	0.53 ± 0.03	0.84 ± 0.13	0.74 ± 0.09	MS, RI
82	24.094	1724	1725	73	60	C13H26O2	Ethyl undecanoate	627–90-7	ND	ND	0.79 ± 0.11	MS, RI
84	24.88	1743	1754	164	91	C10 H12 O2	Ethyl phenylacetate	101–97-3	10.90 ± 0.32	13.77 ± 0.35	9.89 ± 1.23	MS, RI
87	25.565	1812	1811	91	104	C10H12O2	Phenethyl acetate	103–45-7	0.61 ± 0.07	0.73 ± 0.11	1.18 ± 0.21	MS, RI
88	25.758	1822	1829	57	145	C14 H28 O2	Octyl hexanoate	4887-30-3	ND	ND	0.39 ± 0.06	MS, RI
92	27.032	1885	1886	91	104	C11 H14 O2	Ethyl 3-phenylpropionate	2021–28-5	5.36 ± 0.34	9.49 ± 0.25	4.88 ± 0.65	MS, RI
96	28.494	1958	1970	57	88	C12H16O2	Butyl phenylacetate	122–43-0	0.22 ± 0.02	ND	ND	MS, RI
100	29.212	1993	1996	73	55	C15H28O2	Lauryl acrylate	2156-97-0	ND	ND	0.07 ± 0.00	MS, RI
104	30.282	2047	2043	101	88	C16 H32 O2	Ethyl myristate	124–06-1	0.11 ± 0.01	2.52 ± 0.23	0.70 ± 0.13	MS, RI
107	31.311	2098	2114	143	185	C12 H22 O4	Diethyl suberate	2050-23-9	ND	ND	0.04 ± 0.00	MS, RI
110	31.558	2112	2079	155	99	C12H27O4P	Tributyl phosphate	126–73-8	0.03 ± 0.00	ND	ND	MS, RI
112	32.189	2147	2151	101	88	C17 H34 O2	Ethyl pentadecanoate	41,114–00-5	ND	0.44 ± 0.03	0.07 ± 0.02	MS, RI
113	32.434	2160	2164	105	104	C14 H20 O2	2-Phenylethyl hexanoate	6290-37-5	ND	1.17 ± 0.14	0.72 ± 0.12	MS, RI
115	34.027	2252	2253	101	88	C18 H36 O2	Ethyl palmitate	628–97-7	1.21 ± 0.10	24.30 ± 1.53	10.70 ± 1.72	MS, RI
116	34.355	2271	2277	55	88	C18 H34 O2	Ethyl 9-hexadecenoate	54,546–22-4	ND	0.36 ± 0.44	ND	MS, RI
120	35.788	2352	2355	101	88	C19H38O2	Ethyl heptadecanoate	14,010–23-2	ND	0.08 ± 0.01	ND	MS, RI
121	35.902	2358	2358	177	149	C12H14O4	Diethyl phthalate	84–66-2	0.04 ± 0.00	ND	ND	MS, RI
123	37.511	2456	2455	101	88	C20H40O2	Ethyl stearate	111–61-5	ND	0.16 ± 0.02	0.14 ± 0.03	MS, RI
125	37.849	2478	2476	69	55	C20 H38 O2	Ethyl oleate	111–62-6	ND	3.42 ± 0.31	1.36 ± 0.08	MS, RI
128	38.586	2526	2524	81	67	C20H36O2	Ethyl linoleate	544–35-4	0.33 ± 0.05	6.14 ± 0.23	0.74 ± 0.05	MS, RI
129	38.685	2532	2536	57	149	C16H22O4	Diisobutyl phthalate	84–69-5	0.33 ± 0.03	ND	0.29 ± 0.03	MS, RI
130	39.582	2589	2591	95	79	C20H34O2	Ethyl linolenate	1191-41-9	ND	0.26 ± 0.02	ND	MS, RI
132	41.052	2686	2683	150	149	C16H22O4	Dibutyl phthalate	84–74-2	0.50 ± 0.08	0.54 ± 0.08	ND	MS, RI
Acids												
30	15.272	1378	1377	68	88	C9H16O2	Ethyl (*E*)-hept-4-enoate	54,340–70-4	0.43 ± 0.03	ND	0.29 ± 0.05	MS, RI
69	21.699	1638	1637	73	60	C4 H8 O2	Butyric acid	107–92-6	2.66 ± 0.13	3.33 ± 0.35	1.32 ± 0.28	MS, RI
78	22.64	1679	1674	87	60	C5H10O2	2-Methylbutanoic acid	116–53-0	0.35 ± 0.04	0.07 ± 0.00	ND	MS, RI
83	24.126	1725	1720	73	60	C5 H10 O2	Valeric acid	109–52-4	2.01 ± 0.26	2.07 ± 0.25	ND	MS, RI
86	25.519	1810	1813	73	57	C6H12O2	4-Methylpentanoic acid	646–07-1	0.03 ± 0.00	ND	ND	MS, RI
90	26.460	1857	1854	73	60	C6 H12 O2	Caproic acid	142–62-1	103.21 ± 2.02	175.33 ± 13.03	52.36 ± 1.39	MS, RI
95	28.447	1955	1950	73	88	C8 H16 O2	2-Ethylhexanoic acid	149–57-5	ND	ND	0.21 ± 0.05	MS, RI
97	28.553	1961	1960	73	60	C7 H14 O2	Heptanoic acid	111–14-8	8.63 ± 0.63	8.64 ± 0.41	3.65 ± 0.57	MS, RI

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Table 4 (*continued*)

		RI		MS						Relative Concentration (mg L^{-1})[e]		
Peak number	RT (min)	RIref[a]	RIcal[b]	Mass	ion[c]	Formula	Compounds[d]	CAS	GJ	TQ	8NJ	Identification
105	30.613	2064	2070	73	60	C8 H16 O2	Octanoic acid	124–07-2	11.96 ± 0.38	14.84 ± 1.58	4.50 ± 0.35	MS, RI
114	32.648	2172	2171	73	60	C9 H18 O2	Nonanoic acid	112–05-0	0.39 ± 0.03	0.48 ± 0.11	0.27 ± 0.05	MS, RI
117	34.533	2281	2280	60	73	C10H20O2	Decanoic acid	334–48-5	0.44 ± 0.04	1.17 ± 0.08	0.61 ± 0.19	MS, RI
122	36.558	2394	2398	60	73	C11H22O2	Undecanoic acid	112–37-8	0.01 ± 0.00	ND	ND	MS, RI
124	37.665	2466	2457	122	105	C7H6O2	Benzoic acid	65–85-0	0.09 ± 0.01	0.11 ± 0.01	ND	MS, RI
126	38.036	2490	2490	60	73	C12 H24 O2	Lauric Acid	143–07-7	0.16 ± 0.06	0.31 ± 0.02	0.33 ± 0.03	MS, RI
131	40.223	2631	2650	104	91	C9 H10 O2	Hydrocinnamic acid	501–52-0	0.07 ± 0.01	ND	0.06 ± 0.01	MS, RI
133	41.280	2701	2704	60	73	C14 H28 O2	Tetradecanoic acid	544–63-8	0.39 ± 0.14	0.48 ± 0.02	0.27 ± 0.01	MS, RI
134	42.346	2774	2779	60	73	C15 H30 O2	Pentadecanoic acid	1002-84-2	0.19 ± 0.11	ND	0.08 ± 0.00	MS, RI
135	44.372	2915	2910	60	73	C16 H32 O2	n-Hexadecanoic acid	57–10-3	8.27 ± 3.81	11.56 ± 1.93	7.79 ± 1.12	MS, RI
136	44.852	2946	2960	83	55	C16H30O2	Palmitoleic acid	373–49-9	0.44 ± 0.15	ND	0.18 ± 0.03	MS, RI
138	46.048		3027	60	73	C17H34O2	Heptadecanoic acid	506–12-7	0.08 ± 0.02	ND	ND	MS
140	48.142		3136	60	73	C18 H36 O2	Octadecanoic acid	57–11-4	3.56 ± 0.95	2.49 ± 0.08	3.09 ± 0.37	MS
141	48.732		3150	69	55	C18H34O2	Oleic acid	112–80-1	0.51 ± 0.08	2.10 ± 0.14	0.44 ± 0.12	MS
143	50.087		3176	82	67	C18H32O2	Linoleic acid	60–33-3	0.26 ± 0.02	2.52 ± 0.14	0.19 ± 0.01	MS
Aldehydes												
3	3.387	912	807	57	58	C3H6O	Propanal	123–38-6	0.19 ± 0.05	ND	ND	MS, RI
6	4.674	984	988	101	73	C8H18O2	1,1-Diethoxybutane	3658-95-5	ND	0.27 ± 0.05	ND	MS, RI
34	15.728	1394	1392	98	57	C9H18O	Nonanal	124–19-6	0.19 ± 0.04	0.39 ± 0.12	0.17 ± 0.01	MS, RI
47	17.525	1465	1461	57	81	C7H10O	2,4-Heptadienal	4313–03-5	0.03 ± 0.01	ND	ND	MS, RI
48	17.608	1468	1468	95	96	C5H4O2	Furfural	98–01-1	0.16 ± 0.05	0.92 ± 0.45	0.29 ± 0.03	MS, RI
55	18.915	1520	1520	106	105	C7 H6 O	Benzaldehyde	100–52-7	0.19 ± 0.04	0.19 ± 0.09	0.05 ± 0.01	MS, RI
57	19.224	1534	1530	171	103	C13 H28 O2	1,1-Diethoxynonane	54,815–13-3	ND	0.32 ± 0.02	0.04 ± 0.01	MS, RI
70	21.743	1640	1646	92	91	C8H8O	Benzeneacetaldehyde	122–78-1	0.31 ± 0.06	ND	ND	MS, RI
85	25.350	1754	1742	134	133	C9H10O	2,4-Dimethylbenzaldehyde	15,764–16-6	0.04 ± 0.00	ND	ND	MS, RI
119	35.592	2341	2042	82	57	C15H30O	Pentadecanal	2765-11-9	0.02 ± 0.00	ND	ND	MS, RI
Alcohols												
29	15.022	1369	1363	69	56	C6H14O	1-Hexanol	111–27-3	6.30 ± 0.61	ND	5.01 ± 0.32	MS, RI
35	15.885	1399	1396	83	59	C8H18O	3-Octanol	589–98-0	0.06 ± 0.00	ND	ND	MS, RI
44	17.387	1460	1461	56	70	C7H16O	n-heptanol	111–70-6	0.67 ± 0.10	ND	ND	MS, RI
50	18.192	1490	1488	83	57	C8H18O	2-Ethylhexan-1-ol	104–76-7	0.76 ± 0.06	ND	ND	MS, RI
63	19.802	1558	1557	55	56	C8H18O	1-Octanol	111–87-5	0.84 ± 0.04	1.09 ± 0.15	0.39 ± 0.08	MS, RI
72	22.142	1657	1658	70	55	C9 H20 O	1-Nonanol	143–08-8	0.16 ± 0.01	0.53 ± 0.06	ND	MS, RI
79	23.016	1695	1690	83	59	C10 H18 O	L-α-Terpineol	10,482–56-1	0.05 ± 0.00	ND	ND	MS, RI
93	27.543	1901	1912	122	91	C8H10O	Phenethyl alcohol	60–12-8	0.03 ± 0.00	0.46 ± 0.03	0.32 ± 0.04	MS, RI
103	30.090	2038	2042	93	69	C15 H26 O	trans-Nerolidol	40,716–66-3	ND	ND	0.08 ± 0.01	MS, RI
127	38.195	2501	2534	83	57	C18H38O	1-Octadecanol	112–92-5	0.02 ± 0.00	ND	ND	MS, RI

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Table 4 (*continued*)

		RI		MS						Relative Concentration (mg L^{-1})[e]		
Peak number	RT (min)	RIref[a]	RIcal[b]	Mass	ion[c]	Formula	Compounds[d]	CAS	GJ	TQ	8NJ	Identification
Aromatic												
7	5.772	1033	1037	92	91	C7 H8	Toluene	108–88-3	4.34 ± 0.61	ND	ND	MS, RI
12	8.078	1120	1122	106	91	C8H10	Ethylbenzene	100–41-4	ND	0.34 ± 0.01	ND	MS, RI
13	8.2358	1126	1123	106	91	C8H10	1,3-Dimethylbenzene	108–38-3	0.05 ± 0.01	ND	ND	MS, RI
14	8.270	1127	1125	106	91	C8H10	p-Xylene	106–42-3	ND	0.34 ± 0.01	1.92 ± 0.16	MS, RI
40	16.436	1422	1420	190	175	C14 H22	1,3-Di-tert-butylbenzene	1014-60-4	0.40 ± 0.04	ND	ND	MS, RI
49	17.948	1481	1480	134	119	C10 H14	1,2,4,5-Tetramethylbenzene	95–93-2	0.05 ± 0.00	ND	ND	MS, RI
64	20.302	1578	1554	129	130	C10H10	1,4-Divinylbenzene	105–06-6	0.04 ± 0.00	ND	ND	MS, RI
68	21.591	1633	1644	101	133	C11H16	Pentamethylbenzene	700–12-9	0.04 ± 0.00	ND	ND	MS, RI
80	23.338	1705	1711	91	103	C12H18O2	Phenylacetaldehyde diethyl acetal	6314-97-2	0.57 ± 0.05	2.84 ± 0.72	1.47 ± 0.25	MS, RI
81	23.741	1715	1707	127	128	C10 H8	Naphthalene	91–20-3	0.74 ± 0.04	0.83 ± 0.08	ND	MS, RI
89	26.096	1839	1845	141	142	C11 H10	2-Methylnaphthalene	91–57-6	0.13 ± 0.02	0.26 ± 0.03	ND	MS, RI
94	28.428	1955	2012	98	156	C12H12	2,6-Dimethylnaphthalene	581–42-0	0.02 ± 0.00	ND	ND	MS, RI
Ketones												
20	13.508	1311	1314	88	58	C8H16O	2-Octanone	111–13-7	ND	ND	0.67 ± 0.04	MS, RI
33	15.653	1391	1386	71	58	C9 H18 O	2-Nonanone	821–55-6	0.55 ± 0.04	0.43 ± 0.72	0.27 ± 0.17	MS, RI
52	18.45	1500	1496	71	58	C10H20O	2-Decanone	693–54-9	ND	0.13 ± 0.01	ND	MS, RI
58	19.097	1528	1518	57	85	C11 H22 O2	4-Undecanone	14,476–37-0	ND	ND	0.06 ± 0.01	MS, RI
65	20.502	1586	1595	138	82	C9 H14 O	Isophorone	78–59-1	0.14 ± 0.03	ND	ND	MS, RI
102	29.708	2018	2019	71	58	C15H30O	2-Pentadecanone	2345-28-0	ND	0.12 ± 0.02	0.02 ± 0.00	MS, RI
111	31.677	2118	2110	71	58	C18 H36 O	6,10,14-Trimethylpentadecan-2-one	502–69-2	ND	0.17 ± 0.03	ND	MS, RI
Phenol												
101	29.546	2010	2008	66	94	C6H6O	Phenol	108–95-2	0.05 ± 0.00	ND	ND	MS, RI
106	30.964	2081	2081	108	107	C7 H8 O	p-Cresol	106–44-5	0.45 ± 0.06	0.02 ± 0.00	0.38 ± 0.04	MS, RI
118	35.095	2314	2315	206	191	C14 H22 O	2,4-Di-tert-butylphenol	96–76-4	5.84 ± 0.68	4.63 ± 0.16	3.62 ± 0.27	MS, RI
Amine												MS, RI
137	45.281		2973	72	59	C12H25NO	Dodecanamide	1120-16-7	ND	0.13 ± 0.01	ND	MS
142	49.7		2858	72	59	C16 H33 N O	Hexadecanamide	629–54-9	5.65 ± 0.64	5.08 ± 0.82	3.93 ± 0.25	MS
144	57.777		3265	72	59	C18 H35 N O	Oleamide	301–02-0	8.47 ± 0.39	21.02 ± 0.78	6.41 ± 0.29	MS
Others												
2	3.319	908	908	58	57	C5H10O	2-Methylbutyraldehyde	96–17-3	0.10 ± 0.01	0.27 ± 0.03	0.28 ± 0.02	MS, RI
24	14.148	1336	1332	97	103	C11 H24 O2	1,1-Diethoxyheptane	688–82-4	0.15 ± 0.02	ND	ND	MS, RI
36	16.019	1404	1405	83	82	C7H13N	Heptanonitrile	629–08-3	0.10 ± 0.01	ND	ND	MS, RI
46	17.475	1463	1221	111	97	C6 H8 S	2,3-Dimethylthiophene	632–16-6	0.06 ± 0.00	ND	ND	MS, RI
54	18.819	1516	1515	76	87	C8 H16 O3	Ethyl 2-hydroxy-4-methylvalerate	10,348–47-7	0.09 ± 0.02	ND	ND	MS, RI
62	19.71	1554	1557	119	161	C15H24	Longifolene	475–20-7	0.04 ± 0.01	0.43 ± 0.13	0.15 ± 0.00	MS, RI
74	22.288	1664	1661	81	85	C10 H12 O	Estragole	140–67-0	0.01 ± 0.00	ND	ND	MS, RI
91	26.705	1869	1857	98	81	C11 H16 O3	Furfuryl hexanoate	39,252–02-3	ND	0.32 ± 0.01	0.27 ± 0.03	MS, RI

(*continued on next page*)

Table 4 (*continued*)

Peak number	RT (min)	RI		MS		Formula	Compounds[d]	CAS	GJ	Relative Concentration (mg L^{-1})[e]		Identification
		RIref[a]	RIcal[b]	Mass	ion[c]					TQ	8NJ	
108	31.39	2102	2109	95	150	C15H26O	Cedrol	77–53-2	ND	ND	0.02 ± 0.00	MS, RI
109	31.95	2134	2139	138	94	C8H10O2	Phenoxyethanol-	122–99-6	ND	ND	0.05 ± 0.00	MS, RI
139	46.798		3058	81	69	C30H50	Supraene	7683-64-9	0.15 ± 0.03	ND	ND	MS

[a] RIcal: The linear retention index on the VF-WAXms column as calculated by Kovats method (Van den Dool & Kratz, 1963).
[b] RIref: RI referenced from https://webbook.nist.gov with the polarity of the column. ND: not detected.
[c] Characteristic ion.
[d] Compounds: Compounds from https://pubchem.ncbi.nlm.nih.gov/.
[e] Relative Concentration (mg L^{-1}) was calculated using the formula:$C = \frac{A_i \times v_0 \times C_0}{A_0 \times m}$, C : *the relative* Where: C is the relative concentration of the volatile compound; C_0 is the concentration of the internal standard (mg·L^{-1}); A_i is the peak area of the volatile compound;A_0 is the peak area of 2-Octanol; V_0 is the volume of 2-Octanol (μL); and m is the volume of strong-aroma Baijiu sample (mL).

This method's reliability was rigorously established by Ma, Zhang, et al. (2024), who documented excellent stability (intra-day and inter-day relative standard deviations ≤20 %), sensitivity (limits of quantification ranging from 0.12 to 240 μg/L), and linearity ($R^2 \geq 0.994$) across 16 representative compounds. Building on this validated foundation, our semi-quantitative approach enabled efficient cross-sample comparison of flavor profiles while retaining the capacity for absolute quantification when reference standards are available.

Among the 46 common aromatic components, 11 (9 acids and 2 esters) have high RI (> 1560). Notably, nonanoic acid has a cheese-like aroma, whereas fatty acids such as octanoic, decanoic, dodecanoic, tetradecanoic, hexadecanoic, octadecanoic, octadecenoic, oleic, and linoleic acids impart a fatty aroma (Guneser & Yuceer, 2011; Dinh et al., 2021). Fatty acids contribute significantly to the pungency of Baijiu and serve as precursors to esters, thereby playing a critical role in the flavor profile (Wu et al., 2023).

Ethyl linoleate and ethyl tetradecanoate, identified as fatty acid esters, contribute complex, rich flavor layers to Baijiu, while ethyl linoleate imparts a fatty aroma and ethyl tetradecanoate offers a fruity note (Li et al., 2019). This finding enriches the understanding of the aroma composition of Luzhou Laojiao and further validates the effectiveness of the DMU-SPME technique in identifying low-volatility compounds. This method provides valuable insights into the common flavor compounds of strong-flavoured Baijiu, offering a deeper understanding of the unique flavor appeal of Baijiu varieties such as Luzhou Laojiao.

The aromatic components of the three samples were effectively differentiated by OPLS-DA (Fig. 5B). The model fitting indices were as follows: $R^2x = 0.998$, $R^2y = 0.996$, and $Q^2 = 0.989$, with R^2 and Q^2 values above 0.5, indicating acceptable model fitting (Yun et al., 2021). After 200 substitution tests (Fig. 5C), the intersection of the Q^2 regression line and the vertical axis was below zero, confirming that the model was not overfitted and validated for aroma identification.

To further investigate the contribution of different aroma components to various strong-aroma Baijiu, 36 unique flavor compounds were identified based on the Variable Importance in Projection (VIP) plot (Fig. 5D), with a VIP threshold of >1.0. These compounds were further analyzed, and their differences in content across the three Baijiu types were visualised using a Sankey diagram (Fig. 5E). Of the 36 unique flavor compounds, 11 (6 esters, 2 ketones, 1 alcohol, 1 acid and 1 amide) had high RI (> 1560).

Further sensory relevance analysis revealed that specific compounds such as the ketone 2-Pentadecanone and the ester pentadecanoic acid ethyl ester contributed to Baijiu's fruity, floral, and ester-like aroma profiles. Additionally, unsaturated fatty acid esters such as ethyl oleate and ethyl alpha-linolenate exhibit distinct fatty aromas, enhancing the complexity and diversity of the flavor profile of Baijiu. These findings demonstrate that the low-volatile dominant substances identified through DMU-SPME are crucial for distinguishing the aromatic characteristics of the different strong-flavoured Baijiu samples.

Compared to previous studies using HS-SPME for concentrated Baijiu analysis (Fan & Qian, 2005), the DMU-SPME technique not only detected short-chain and medium-chain fatty acids and common esters but also revealed a greater number of long-chain fatty acids and unsaturated fatty acid esters. This suggests that DMU-SPME might offer superior sensitivity and broader detection range than HS-SPME for analysing complex flavor components in strongly flavoured Baijiu. Thus, it represents a promising and innovative tool for flavor analysis and quality control of Baijiu.

4. Conclusions

The integration of DMU-SPME with variable-temperature extraction establishes a novel analytical platform for comprehensive volatilomics profiling of strong-aroma Baijiu. Through systematic optimization of key parameters (temperature programming, salt addition, fiber immersion depth), this approach demonstrated exceptional efficacy in isolating challenging low-volatility constituents (e.g., unsaturated fatty acid esters and long-chain fatty acids). DMU-SPME with variable-temperature extraction identified 103 substances, exceeding headspace SPME (HS-SPME) by 32 and immersion SPME (DI-SPME) by 25. It uniquely extracted 19 compounds (7 esters, 8 acids, 1 alcohol, 2 aldehydes, 1 hydrocarbon). Critically, analysis of three commercial grades revealed 46 common characteristic flavor compounds (11 with RI > 1560) and 36 unique markers (VIP > 1.0) across three grades of strong-aroma Baijiu (11 with RI > 1560), providing unprecedented resolution of shared and distinct low-volatility components. These findings fundamentally advance extraction science by demonstrating how dual-mode/temperature protocols overcome competitive adsorption effects that limit single-mode SPME. Consequently, DMU-SPME with variable-temperature extraction emerges as a robust, high-coverage tool for aroma profile assessment directly linked to Baijiu quality grading. Future studies will expand method validation across diverse spirit products and integrate GC-olfactometry to establish quantitative structure-sensory relationships for the newly identified compounds.

CRediT authorship contribution statement

Xue Wang: Writing – review & editing, Writing – original draft, Visualization, Investigation, Data curation, Conceptualization. **Xian-Bing Xu:** Writing – review & editing, Methodology, Funding acquisition,

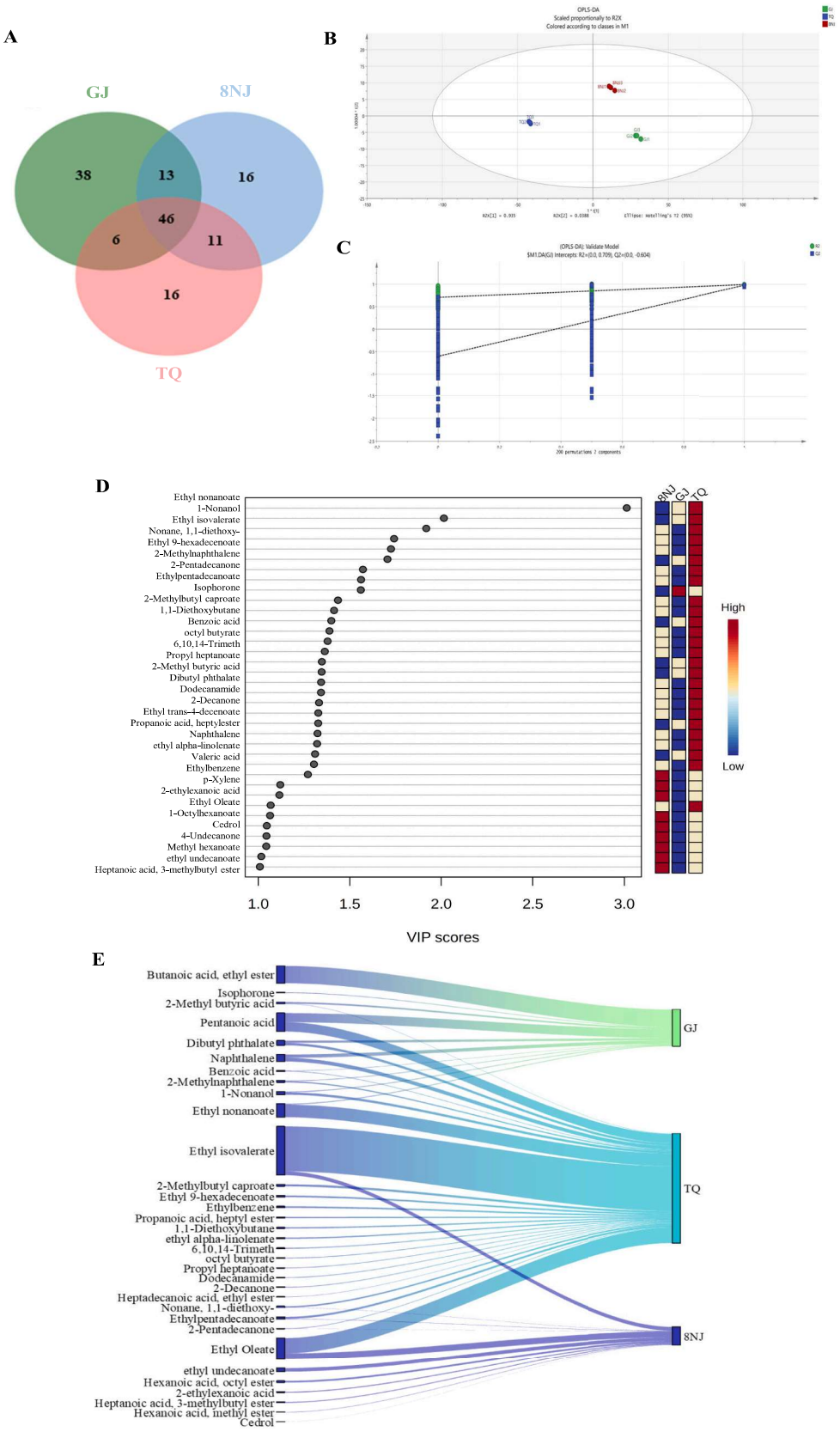


Fig. 5. (A) Venn diagram of the number of compounds extracted by GJ, 8NJ and TQ. (B) OPLS-DA score plot based on the common aroma compounds of GJ, 8NJ and TQ. (C) Permutation test of permutation test of OPLS-DA model on the common aroma compounds of GJ, 8NJ, and TQ. (D) VIP scores of the differential aroma substances extracted by GJ, 8NJ, and TQ. The colored boxes on the right indicate the peak area of the corresponding compounds. Red colour indicates high and blue colour indicates low. (E) The Sankey diagram illustrates the differences content distribution across GJ, 8NJ, and TQ. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Conceptualization. **Ling Zhang:** Validation. **Qi-Yue Zhao:** Validation. **Jin-Feng Pan:** Methodology. **Yi-feng Fan:** Resources. **Xuan-Peng Wang:** Writing – review & editing, Supervision, Resources, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

All authors would like to thank the Liaoning Specialty Food Preparedness Key Technology Research and Demonstration Project (Grant No. 2023JH1/11200001).

Data availability

Data will be made available on request.

References

Câmara, J., Marques, J., Perestrelo, R., Rodrigues, F., Oliveira, L., Andrade, P., & Caldeira, M. (2007). Comparative study of the whisky aroma profile based on headspace solid phase microextraction using different fibre coatings. *Journal of Chromatography A, 1150*(1–2), 198–207. https://doi.org/10.1016/j.chroma.2006.09.014

Chen, E., Yang, F., Ma, Z., Geng, X., Li, Q., Zheng, F., & Sun, B. (2024). Analysis of aroma compounds in different grades of Luzhou Laojiao strong-aroma baijiu by molecular sensory science. *Journal of Food Composition and Analysis, 135*, Article 106607. https://doi.org/10.1016/j.jfca.2024.106607

Cortada, C., Vidal, L., & Canals, A. (2011). Determination of nitroaromatic explosives in water samples by direct ultrasound-assisted dispersive liquid–liquid microextraction followed by gas chromatography–mass spectrometry. *Talanta, 85*(5), 2546–2552. https://doi.org/10.1016/j.talanta.2011.08.011

David, F., & Sandra, P. (2007). Stir bar sorptive extraction for trace analysis. *Journal of Chromatography A, 1152*(1–2), 54–69. https://doi.org/10.1016/j.chroma.2007.01.032

Deuscher, Z., Gourrat, K., Repoux, M., Boulanger, R., Labouré, H., & Le Quéré, J.-L. (2020). Key aroma compounds of dark chocolates differing in organoleptic properties: A GC-O comparative study. *Molecules, 25*(8), 1809. https://doi.org/10.3390/molecules25081809

Duru, M. E., Cakir, A., & Harmandar, M. (2002). Composition of the volatile oils isolated from the leaves of Liquidambar orientalis mill. var. orientalis and *L. orientalis* var. integriloba from Turkey. *Flavour and Fragrance Journal, 17*(2), 95–98. https://doi.org/10.1002/ffj.1050

Fan, W., & Qian, M. C. (2005). Headspace solid phase microextraction and gas chromatography−olfactometry dilution analysis of young and aged Chinese "Yanghe Daqu" liquors. *Journal of Agricultural and Food Chemistry, 53*(20), 7931–7938. https://doi.org/10.1021/jf051011k

Fiorini, D., Pacetti, D., Gabbianelli, R., Gabrielli, S., & Ballini, R. (2015). A salting out system for improving the efficiency of the headspace solid-phase microextraction of short and medium chain free fatty acids. *Journal of Chromatography A, 1409*, 282–287. https://doi.org/10.1016/j.chroma.2015.07.051

Guneser, O., & Yuceer, Y. K. (2011). Characterisation of aroma-active compounds, chemical and sensory properties of acid-coagulated cheese: Circassian cheese. *International Journal of Dairy Technology, 64*(4), 517–525. https://doi.org/10.1111/j.1471-0307.2011.00703.x

Hong, J., Huang, H., Zhao, D., Sun, J., Huang, M., Sun, X., & Sun, B. (2023). Investigation on the key factors associated with flavor quality in northern strong aroma type of baijiu by flavor matrix. *Food Chemistry, 426*, Article 136576. https://doi.org/10.1016/j.foodchem.2023.136576

Hong, J., Wang, J., Zhang, C., Zhao, Z., Tian, W., Wu, Y., Chen, H., Zhao, D., & Sun, J. (2021). Unraveling variation on the profile aroma compounds of strong aroma type of baijiu in different regions by molecular matrix analysis and olfactory analysis. *RSC Advances, 11*(53), 33511–33521. https://doi.org/10.1039/d1ra06073b

Kustrin, S., Gegechkori, V., Kobakhidze, T., & Morton, D. (2023). *Solid-Phase Microextraction Techniques and Application in Food and Horticultural Crops*. https://doi.org/10.3390/molecules28196880

Le Bizec, B., Pinel, G., & Antignac, J.-P. (2009). Options for veterinary drug analysis using mass spectrometry. *Journal of Chromatography A, 1216*(46), 8016–8034. https://doi.org/10.1016/j.chroma.2009.07.007

Li, H., Qin, D., Wu, Z., Sun, B., Sun, X., Huang, M., & Zheng, F. (2019). Characterization of key aroma compounds in Chinese Guojing sesame-flavor baijiu by means of molecular sensory science. *Food Chemistry, 284*, 100–107. https://doi.org/10.1016/j.foodchem.2019.01.102

Li, T., Wang, J., Xu, B., Sun, H., Liu, H., Wang, D., Shen, Y., Wu, J., et al. (2023). Comparative analysis of the differences among Langya flavor baijiu and strong and soy sauce flavor baijiu by targeted flavor analysis. *Journal of Food Composition and Analysis, 122*, Article 105479. https://doi.org/10.1016/j.jfca.2023.105479

Liu, Q.-R., Lin, X.-L., Lu, Z.-M., Chai, L.-J., Wang, S.-T., Shi, J.-S., … Xu, Z.-H. (2024). Influence on the volatilization of ethyl esters: Nonnegligible role of long-chain fatty acids on baijiu flavor via intermolecular interaction. *Food Chemistry, 436*, Article 137731. https://doi.org/10.1016/j.foodchem.2023.137731

Liu, Y., Huang, Y., Chen, G., Huang, J., Zheng, J., Xu, J., Liu, S., Qiu, J., et al. (2018). A graphene oxide-based polymer composite coating for highly-efficient solid phase microextraction of phenols. *Analytica Chimica Acta, 1015*, 20–26. https://doi.org/10.1016/j.aca.2018.02.034

Liu, Y., Qiao, Z., Zhao, Z., Wang, X., Sun, X., Han, S., & Pan, C. (2022). Comprehensive evaluation of Luzhou-flavor liquor quality based on fuzzy mathematics and principal component analysis. *Food Science & Nutrition, 10*(6), 1780–1788. https://doi.org/10.1002/fsn3.2796

Ma, Y.-J., Li, P., Zhu, B.-W., Du, M., & Xu, X.-B. (2024). Comprehensive determination of fatty acids in real samples without derivatization by DMU-SPME-GC methods. *Food Research International, 195*, Article 114986. https://doi.org/10.1016/j.foodres.2024.114986

Ma, Y.-J., Zhang, L., Zhu, B.-W., Du, M., & Xu, X.-B. (2024). Theory and protocol of dual mode unity solid-phase microextraction. *Talanta, 270*, Article 125573. https://doi.org/10.1016/j.talanta.2023.125573

Ma, Y.-J., Zhou, T., Jiang, W., Zhu, B.-W., Du, M., & Xu, X.-B. (2023). Balanced extraction of volatile and semi-volatile compounds by dynamic linked position unity solid-phase microextraction. *Food Chemistry, 407*, Article 135160. https://doi.org/10.1016/j.foodchem.2022.135160

Merkle, S., Kleeberg, K. K., & Fritsche, J. (2015). Recent developments and applications of solid phase microextraction (SPME) in food and environmental analysis—A review. *Chromatography, 2*(3), 293–381. https://doi.org/10.3390/chromatography2030293

Miao, Z., Hao, H., Yan, R., Wang, X., Wang, B., Sun, J., & Sun, B. (2022). Individualization of Chinese alcoholic beverages: Feasibility towards a regulation of organic acids. *LWT, 172*, Article 114168. https://doi.org/10.1016/j.lwt.2022.114168

Mirnaghi, F. S., Goryński, K., Rodriguez-Lafuente, A., Boyacı, E., Bojko, B., & Pawliszyn, J. (2013). Microextraction versus exhaustive extraction approaches for simultaneous analysis of compounds in wide range of polarity. *Journal of Chromatography A, 1316*, 37–43. https://doi.org/10.1016/j.chroma.2013.09.084

Monforte, A. R., Martins, S. I., & Ferreira, A. C. S. (2021). Phenylacetaldehyde real-time release kinetics in wine like model solutions. *Food Chemistry, 364*, Article 128948. https://doi.org/10.1016/j.foodchem.2020.128948

Risticevic, S., Lord, H., Gorecki, T., Arthur, C. L., & Pawliszyn, J. (2010). Protocol for solid-phase microextraction method development. *Nature Protocols, 5*(1), 122–139. https://doi.org/10.1038/nprot.2009.179

Rocha, S., Ramalheira, V., Barros, A., Delgadillo, I., & Coimbra, M. A. (2001). Headspace solid phase microextraction (SPME) analysis of flavor compounds in wines. Effect of the matrix volatile composition in the relative response factors in a wine model. *Journal of Agricultural and Food Chemistry, 49*(11), 5142–5151. https://doi.org/10.1021/jf010566m

Ruvalcaba, J. E., Durán-Guerrero, E., Barroso, C. G., & Castro, R. (2019). Development of a stir bar sorptive extraction method to study different beer styles volatile profiles. *Food Research International, 126*, Article 108680. https://doi.org/10.1016/j.foodres.2019.108680

Salum, P., Erbay, Z., Kelebek, H., & Selli, S. (2017). Optimization of headspace solid-phase microextraction with different fibers for the analysis of volatile compounds of white-brined cheese by using response surface methodology. *Food Analytical Methods, 10*, 1956–1964. https://doi.org/10.1007/s12161-016-0774-1

Tachihara, T., Hashimoto, H., Ishizaki, S., Komai, T., Fujita, A., Ishikawa, M., & Kitahara, T. (2006). Microbial resolution of 2-methylbutyric acid and its application to several chiral flavour compounds. In *Developments in food science* (pp. 97–100). Elsevier. https://doi.org/10.1016/S0167-4501(06)80023-3.

Van Den Dool, H., & Kratz, P. D. (1963). A generalization of the retention index system including linear temperature programmed gas-liquid partition chromatography. *Journal of chromatography., 11*, 463–471. https://doi.org/10.1016/S0021-9673(01)80947-X

Wang, Z., Wang, S., Liao, P., Chen, L., Sun, J., Sun, B., Zhao, D., Wang, B., & Li, H. (2022). HS-SPME combined with GC-MS/O to analyze the flavor of strong aroma Baijiu Daqu. *Foods, 11*(1), 116. https://doi.org/10.3390/foods11010116

Wu, Y., Chen, H., Huang, H., Chen, F., Hong, J., Zhao, D., Zhang, C., Zhao, Z., et al. (2023). Revelation for the influence mechanism of long-chain fatty acid ethyl esters on the baijiu quality by multicomponent chemometrics combined with modern flavor sensomics. *Foods, 12*(6), 1267. https://doi.org/10.3390/foods12061267

Yamazaki, Y., Sasaki, T., Ochiai, N., Sasamoto, K., Michihata, T., & Koyanagi, T. (2024). Target profiling of aroma compounds in three types of Japanese sake using solvent-assisted stir bar sorptive extraction with gas chromatography–mass spectrometry. *Journal of Food Composition and Analysis, 133*, Article 106441. https://doi.org/10.1016/j.jfca.2024.106441

Yan, S., Wang, S., Wei, G., & Zhang, K. (2015). Investigation of the main parameters during the fermentation of Chinese Luzhou-flavour liquor. *Journal of the Institute of Brewing, 121*(1), 145–154. https://doi.org/10.1002/jib.193

Yang, C., Wang, J., & Li, D. (2013). Microextraction techniques for the determination of volatile and semivolatile organic compounds from plants: A review. *Analytica Chimica Acta, 799*, 8–22. https://doi.org/10.1016/j.aca.2013.07.069

Yun, J., Cui, C., Zhang, S., Zhu, J., Peng, C., Cai, H., Yang, X., & Hou, R. (2021). Use of headspace GC/MS combined with chemometric analysis to identify the geographic origins of black tea. *Food Chemistry, 360*, Article 130033. https://doi.org/10.1016/j.foodchem.2021.130033

Zhang, M., Wu, X., Mu, D., Xu, B., Xu, X., Chang, Q., & Li, X. (2021). Profiling the influence of physicochemical parameters on the microbial community and flavor substances of zaopei. *Journal of the Science of Food and Agriculture, 101*(15), 6300–6310. https://doi.org/10.1002/jsfa.11299

Zhang, W., Hu, Z., Yi, C., Shan, Y., & Fang, Z. (2025). Improving rice fermentation flavor: Screening and evaluation of two endogenous flavor-producing strains. *Food Bioscience, 68*, Article 106433. https://doi.org/10.1016/j.fbio.2025.106433

Zhang, X., Wang, C., Wang, L., Chen, S., & Xu, Y. (2020). Optimization and validation of a head space solid-phase microextraction-arrow gas chromatography-mass spectrometry method using central composite design for determination of aroma compounds in Chinese liquor (Baijiu). *Journal of Chromatography A, 1610*, Article 460584. https://doi.org/10.1016/j.chroma.2019.460584

Zhu, H., Li, L., Ma, Y., Luo, H., El-Sappah, A. H., Liu, Y., & Sun, Y. (2025). Volatilomics and Lipidomics revealed flavoring mechanism in baijiu brewed from diverse Sorghum varieties. *Food Chemistry, 144503*. https://doi.org/10.1016/j.foodchem.2025.144503